

Introduction to Machine Learning

Machine learning is a branch of artificial intelligence that enables computers to learn from data without being explicitly programmed.

The field was pioneered by Arthur Samuel in 1959.

There are three main paradigms: supervised learning where models learn from labeled examples, unsupervised learning where models find hidden patterns, and reinforcement learning where agents learn from rewards.

Deep Learning and Neural Networks

Deep learning uses neural networks with multiple layers. CNNs excel at image recognition. RNNs handle sequential data. Transformers have revolutionized NLP since 2017.

Applications include image classification, speech recognition, natural language processing, and autonomous vehicles.

Applications of Machine Learning

In healthcare ML models detect diseases from medical images.

In finance algorithms detect fraudulent transactions in real time.

Recommendation systems use collaborative filtering to suggest content.

Self driving cars use computer vision and reinforcement learning.